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Costas Arrays with Small Number of Cross-Coincidences

A simple method is described of obtaining two or more Costas arrays with a maximum of one mutual hit or cross-coincidence, assuming that the vertical components of their relative shifts are appropriately bounded. This result can be useful for designing multiple Costas waveforms with small cross-ambiguities, if the potential Doppler shifts are restricted to a small portion of the waveform bandwidth.

I. INTRODUCTION

A problem of considerable interest to the radar and sonar communities is to find a class of transmit waveforms with nearly ideal range-Doppler auto-ambiguity and cross-ambiguity characteristics. The auto-ambiguity function should be "thumbtack-like", that is, outside of a narrow mainlobe around the origin of the range-Doppler plane its values are uniformly small, while the cross-ambiguity function should be small everywhere. Costas waveforms are a class of frequency-hopping waveforms that have nearly ideal auto-ambiguity functions [3]. We examine the problem of finding multiple Costas waveform patterns with small cross-ambiguities when the allowable Doppler shift is constrained. We begin with some definitions and a more precise problem statement.

Costas waveforms are constructed from diagrams called Costas arrays.

DEFINITION 1. *Costas Array.* Consider an array consisting of n rows and n columns, which has exactly one mark in each row and each column. Such an array is called an $n \times n$ Costas array if the coincidence function $C(r,s)$ satisfies

$$C(r,s) \leq 1$$

for all integer pairs $(r,s) \neq (0,0)$ with $|r| \leq n-1$, $|s| \leq n-1$.

The coincidence function $C(r,s)$ is defined as the number of the coincidences of marks between the original array and its translation. The parameters r and s define the amount of translation; r is the number of integer shifts to the right or left (translated by columns), and s is the number of integer shifts up or down (translated by rows). This is illustrated in Fig. 1, for $n = 6$, $r = 2$, $s = 3$. Note $C(2,3) = 1$.

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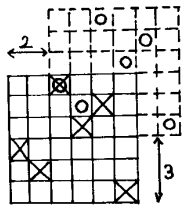


Fig. 1. 6 x 6 Costas array and translation by (2,3).
X = original Costas array, O = translated Costas array, $C(2,3) = 1$.

Each row of an array corresponds to a frequency channel and each column to a subpulse time segment of the waveform. The number of subpulses and frequency channels is n. The coincidence function may be regarded as a discrete analogue of the (unnormalized) ambiguity function. Similarly, a discrete analogue of the cross-ambiguity function is called the cross-coincidence function and is defined by the following.

DEFINITION 2. *Cross-Coincidence Function.* Let A and B be $n \times n$ arrays. For each integer pair (r,s), their cross-coincidence $C_{AB}(r,s)$ is defined as the number of coincidences of marks between A and a translated version of B, which has been shifted to the right by r and up by s.

To obtain two or more Costas waveforms with small cross-ambiguities, it is natural to seek a solution to the following problem.

Problem 1. Find two $n \times n$ Costas arrays, A and B, such that $C_{AB}(r,s) \leq 1$ for all integer pairs with $|r| \leq n-1$, $|s| \leq n-1$.

Unfortunately, it is not possible to find two such arrays if $n \geq 4$ [4, 7]. However, in many applications, the potential Doppler shifts of the echoes can be bounded, due to physical considerations, to a small portion of the total bandwidth of each waveform. Then the cross-ambiguity function needs to be small only for this subset of physically meaningful Doppler values. In terms of cross-coincidences between the arrays, this corresponds to relaxing the requirements of Problem 1 by restricting the range of vertical shifts s. One can thus pose a constrained version of Problem 1 as Problem 2.

Problem 2. Given an integer m $(0 \leq m \leq n-1)$, find two $n \times n$ Costas arrays A, B such that $C_{AB}(r,s) \leq 1$ for all integer pairs with $|r| \leq n-1$, $|s| \leq m$.

For convenience in developing the solution to Problem 2, we consider the following modification, where the limits on r and s are interchanged.

Problem 3. Given an integer m $(0 \leq m \leq n-1)$, find two $n \times n$ Costas arrays A, B such that $C_{AB}(r,s) \leq 1$ for all integer pairs with $|r| \leq m$, $|s| \leq n-1$.

By exchanging the roles of rows and columns of the arrays, it is easy to see that Problem 2 is solved once Problem 3 is solved. In fact, one need only to take the solution arrays to Problem 3 and rotate them either clockwise or counter-clockwise by 90 deg to obtain solution arrays to Problem 2.

It is shown that Problem 3 can be solved provided that 1) $n = p-1$, where p is prime, and 2) $m \leq (n-2)/2$. In fact, it is shown that q Costas arrays with the desired cross-coincidence properties can be found if $m \leq (n-q)/q$. This result is stated (Proposition 1) and proven in Section II. The Proof of Proposition 1 provides a procedure to obtain solution arrays, which is summarized in Section III.

A number of related papers have been published in this area. In [2, 8, 9, 10], Titlebaum and his associates discuss tradeoffs between auto- and cross-ambiguity properties of a wide variety of hop codes including their extended quadratic congruence (EQC) codes. No Doppler constraints are imposed in their work. Mersereau and Seay [6, p. 574] discuss a Doppler constraint problem similar to ours. However, their hopping patterns are $(n+1) \times n$ arrays rather than $n \times n$ arrays. Thus not every frequency channel is used in each waveform, and the maximum number of mutual hits is two. In contrast, only Costas codes are considered in our work.

II. MAIN RESULT

PROPOSITION 1. *Let $n = p-1$, where p is a prime number, and let m and q be positive integers such that*

$$(m+1)q \leq n. \tag{1}$$

Then one can find a group of q distinct $n \times n$ Costas arrays such that any two of them, say A and B, satisfy

$$C_{AB}(r,s) \leq 1 \tag{2}$$

for all integer pairs (r,s) with

$$|r| \leq m; \qquad |s| \leq n-1. \tag{3}$$

PROOF. An $n \times n$ Costas array is represented by an integer sequence, $a_1, a_2, \ldots, a_n$ chosen from the set $\{1,2,\ldots,n\}$ in such a way that the ith column of the array has a mark in the a_ith row. Note that if two $n \times n$ arrays, A and B, are represented by the sequences $a_1,\ldots,a_n$ and $b_1,\ldots,b_n$, respectively, then the condition

$$C_{AB}(r,s) \geq 2 \tag{4}$$

implies that for some integers k and h,

$$a_k - a_{k+h} = b_{k-r} - b_{k+h-r} \tag{5}$$

where $1 \leq k, h \leq n-1$. In fact (5) simply means that the marks of array A on the kth and $(k+h)$th

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columns overlap with the marks of array B on the $(k-r)$th and $(k+h-r)$th columns, respectively, when B is shifted to the right by r and up by some amount s.

With these preliminary remarks, we now begin the construction of the solution. Let α be a primitive element of the Galois field GF(p) (see [11] for the basic properties of Galois fields). Then by definition, in arithmetic modulo p, the powers $\alpha, \alpha^2, \alpha^3, \ldots, \alpha^n$ are all distinct and exhaust all nonzero elements $1,2,\ldots,n$ of GF(p). A primitive element always exists in any Galois field. It is also known that $\alpha^n = \alpha^{p-1} = 1 \pmod p$. Define integers

$$d_j = (m+1)(j-1)+1, \qquad j = 1,\ldots,q \tag{6}$$

and $n \times n$ arrays

$$A_j \sim [\alpha^{d_j}, \alpha^{d_j+1}, \ldots, \alpha^{d_j+n-1}], \qquad j = 1,\ldots,q \tag{7}$$

where the integer q satisfies (1) and the powers in (7) are computed modulo p to yield numbers of the set $\{1,2,\ldots,n\}$. It is well known that each A_j is a Costas array [5, p. 1154]. In particular, A_1 is known as a Welch array.

Also it is easy to see that each A_j is a cyclic shift of A_1, as illustrated in the example of Section III.

Now we claim that the cross-coincidence function between arrays A_i and A_j satisfies

$$C_{A_iA_j}(r,s) \le 1 \tag{8}$$

for all $|r| \le m$, $|s| \le n-1$, and $1 \le i, j \le q$, $i \ne j$. Let us assume otherwise. Suppose

$$C_{A_iA_j}(r,s) \ge 2, \qquad \text{for some} \quad |r| \le m \text{ and } s. \tag{9}$$

Then from (4) and (5), there exist integers k and h, $1 \le k, h \le n-1$, such that the following equality holds:

$$\alpha^{d_i+k-1} - \alpha^{d_i+k+h-1} = \alpha^{d_j+k-1-r} - \alpha^{d_j+k+h-1-r} \pmod p.$$

Simplifying, one obtains

$$\alpha^{d_i+k-1}(1-\alpha^h) = \alpha^{d_j+k-1-r}(1-\alpha^h) \pmod p. \tag{10}$$

Since $1 \le h \le n-1$, then by the definition of a primitive element,

$$1-\alpha^h \ne 0 \pmod p.$$

Upon cancelling this and other nonzero factors in modulo p arithmetic, (10) may be rewritten as

$$\alpha^{d_i} = \alpha^{d_j-r} \pmod p.$$

Again by the basic property of a primitive element this last equality implies that

$$d_i = d_j - r \pmod n. \tag{11}$$

From the assumption (1) and the definition of the d_j (6), this is impossible if $i \ne j$ and $|r| \le m$. To see why this is so, note that from (6),

$$d_i - d_j = (m+1)(i-j).$$

Using the conditions $i \ne j$ and $1 \le i, j \le q$, one obtains

$$m+1 \le |d_i - d_j| \le (m+1)(q-1).$$

Applying (1) to this

$$m+1 \le |d_i - d_j| \le n-(m+1).$$

If $|r| \le m$, it follows that

$$1 \le |d_i - d_j + r| \le n-1.$$

Finally, this implies that

$$d_i - d_j + r \ne 0 \pmod n.$$

This completes the proof that (11) cannot be true if $i \ne j$ and $|r| \le m$. Thus (9) has resulted in a contradiction, and (8) must hold. We have now obtained q distinct Costas arrays with small cross-coincidences for restricted r values.

III. SOLUTION STEPS

The solution to Problem 2 based on Proposition 1 can be summarized as follows. It is assumed that $n = p-1$, where p is a prime number and $(m+1)q \le n$ for some positive integer q.

Step 1. Construct an $n \times n$ Welch Costas array using a primitive element of Galois field GF(p). Call it A_1.

Step 2. Extend the above Welch Costas array horizontally to the right by periodic extension with period n.

Step 3. From this periodically extended array, take an $n \times n$ subarray starting at the $m+2$–nd column and call it A_2. Similarly A_j is defined to be the $n \times n$ subarray starting at the $(m+1)(j-1)+1$st column, for $j = 1,\ldots,q$.

Step 4. $A_1,\ldots,A_q$ provide a solution to Problem 3. Therefore, by rotating these arrays clockwise or counter-clockwise by 90 deg, a solution to Problem 2 is obtained.

EXAMPLE

In Fig. 2, an example of the construction of Proposition 1 is presented. Here $n = 6$, $p = 7$, $m = 2$, $q = 2$. The primitive element of GF(7) chosen here is $\alpha = 3$. Thus A_1 is represented by a sequence of six numbers 3, 2, 6, 4, 5, 1 corresponding to $3, 3^2, 3^3, 3^4, 3^5, 3^6$ modulo 7. The upper 6×12 array in Fig. 2 represents the periodic extension referred to at Step 2. A_2 is obtained by taking the 6×6 subarray starting at the column $m+2 = 4$. These two arrays have a maximum of one cross-coincidence when the horizontal

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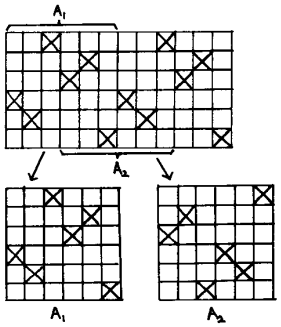


Fig. 2. Example with $n = 6$ ($p = 7$), $m = 2$, $q = 2$. $\alpha = 3$ is a primitive element of GF(7). A_1: 3, 2, 6, 4, 5, 1; A_2: 4, 5, 1, 3, 2, 6.

component of their relative shifts is constrained to be two or less. By simultaneously rotating A_1 and A_2 by 90 deg, a pair of arrays with cross-coincidence at most one is obtained, if the vertical shift is constrained to be two or less.

IV. SUMMARY

The main motivation for this work is the construction of waveforms for radar and sonar applications that have ideal range-Doppler auto- and cross-ambiguity properties. For frequency-hopping signals, this problem leads one to search for multiple arrays (hopping patterns) with a small number of self and mutual hits.

In this work, only Costas codes are considered. Thus, the number of self hits of each waveform is already optimized due to the properties of Costas codes. By introducing a Doppler constraint which may arise from physical considerations, a simple method is presented of constructing multiple Costas codes with at most one mutual hit. The method only requires knowledge of a primitive element in a Galois field. There is a tradeoff between the number of Costas codes obtained by our method and the Doppler constraint. The relation between the number q of the codes and the Doppler constraint m is given by (1). As the range of allowed Doppler shifts (m) increases, the number of Costas codes with small cross-coincidences (q) decreases.

It should be noted, however, that the waveform design procedure of this paper is restricted to the array (or diagram) level. In particular, when the adjacent frequency channels are not sufficiently separated, interferences occur between these channels, which affects the cross-ambiguity properties of the waveforms. Thus the procedure presented here should be regarded as a useful and convenient first step in an overall design process.

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REFERENCES

[1] Chang, W. (1987)
A remark on the definition of Costas arrays.
Proceedings of the IEEE, **75**, 4 (Apr. 1987), 522–523.

[2] Bellegarda, J., and Titlebaum, E. (1987)
Time frequency hop codes based upon extended quadratic congruences.
(To appear in *IEEE Transactions on Aerospace and Electronic Systems*).

[3] Costas, J. (1984)
A study of a class of detection waveforms having nearly ideal range-Doppler ambiguity properties.
Proceedings of the IEEE, **72**, 8 (Aug. 1984), 996–1009.

[4] Freedman, A., and Levanon, N. (1985)
Any $N \times N$ Costas signals must have at least one common ambiguity sidelobe if $N > 3$—A Proof.
Proceedings of the IEEE, **73**, 10 (Oct. 1985), 1530–1531.

[5] Golomb, S. W., and Taylor, H. (1984)
Construction and properties of Costas arrays.
Proceedings of the IEEE, **72**, 9 (Sept. 1984), 1143–1163.

[6] Mersereau, R., and Seay, T. (1981)
Multiple access frequency hopping patterns with low ambiguity.
IEEE Transactions on Aerospace and Electronic Systems, AES-17, 4 (July 1981), 571–578.

[7] Taylor, H. (1984)
Nonattacking rooks with distinct differences.
Technical Report 84-03-02, Communication Science Institute, University of Southern California, Los Angeles, Calif.

[8] Titlebaum, E. (1981)
Time-frequency hop signals Part 1: Coding based upon the theory of linear congruences.
IEEE Transactions on Aerospace and Electronic Systems, AES-17, 4 (July 1981), 490–493.

[9] Titlebaum, E., and Bellegarda, J. (1987)
Bounds on auto- and cross-ambiguity functions for Costas arrays. (Preprint)

[10] Titlebaum, E., and Sibul, L. (1981)
Time-frequency hop signals Part 2: Coding based upon quadratic congruences.
IEEE Transactions on Aerospace and Electronic Systems, AES-17, 4 (July 1981), 494–499.

[11] McEliece, R. J. (1987)
Finite Fields for Computer Scientists and Engineers.
Kluwer Academic Publishers, 1987.

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